AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	ADP South Dakota, SD -- station KFSD, America/Chicago
Building	OSM 545895086 ADP South Dakota
Facade gap	not applicable -- one building, no second facade
Weather record	43,797 real hourly records from KFSD
Plume physics	NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-12-05 (crossing)
Plant limit	18.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 10 of 24 hours with 1 mode change. The reactive incumbent took 13 hours -- 3 more -- but declared 1 unsafe hour against the agent's 0. 12 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	10 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	13 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s) 12 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	0.810	18.0	0.560	1.133
01	mechanical	yes	switch budget	1.110	18.0	1.110	1.053
02	mechanical	yes	switch budget	0.840	18.0	0.560	0.990
03	mechanical	yes	switch budget	0.840	18.0	0.560	0.915

04	mechanical	yes	switch budget	1.390	18.0	1.110	0.865
05	mechanical	yes	switch budget	0.585	18.0	0.000	0.813
06	mechanical	yes	switch budget	0.530	18.0	-0.610	0.862
07	mechanical	yes	switch budget	1.915	18.0	-0.610	1.430
08	mechanical	yes	switch budget	3.335	18.0	0.560	1.926
09	mechanical	yes	switch budget	4.695	18.0	1.670	2.219
10	mechanical	no	dry-bulb	25.250	18.0	42.220	2.091
11	mechanical	yes	-	7.500	18.0	6.110	1.941
12	mechanical	yes	-	7.500	18.0	6.110	1.729
13	mechanical	no	dry-bulb	26.945	18.0	6.110	1.317
14	FREE	yes	-	7.500	18.0	4.440	1.161
15	FREE	yes	-	5.830	18.0	2.220	1.047
16	FREE	yes	-	4.445	18.0	0.560	1.026
17	FREE	yes	-	2.470	18.0	-0.610	1.042
18	FREE	yes	-	0.830	18.0	-1.110	1.290
19	FREE	yes	-	-0.525	18.0	-1.110	1.161
20	FREE	yes	-	-1.415	18.0	-2.220	1.654
21	FREE	yes	-	-1.665	18.0	-2.220	1.606
22	FREE	yes	-	-1.665	18.0	-2.220	1.363
23	FREE	yes	-	-2.220	18.0	-2.220	1.059

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 0.810 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 1.110 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 0.840 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 0.840 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 1.390 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 0.585 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 0.530 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 1.915 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 3.335 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.695 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.250 C against a 18.0 C limit, so it fails by 7.250 C. It would take a limit 7.250 C higher, or a bound 7.250 C tighter, to change this hour.

11:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.500 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

12:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.500 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.945 C against a 18.0 C limit, so it fails by 8.945 C. It would take a limit 8.945 C higher, or a bound 8.945 C tighter, to change this hour.

14:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.500 C, which is 10.500 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.161 C, and the margin is measured, not chosen: 1.161 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

15:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 5.830 C, which is 12.170 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.047 C, and the margin is measured, not chosen: 1.047 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

16:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 4.445 C, which is 13.555 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.026 C, and the margin is measured, not chosen: 1.026 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

17:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 2.470 C, which is 15.530 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.042 C, and the margin is measured, not chosen: 1.042 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 0.830 C, which is 17.170 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.290 C, and the margin is measured, not chosen: 1.290 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -0.525 C, which is 18.525 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.161 C, and the margin is measured, not chosen: 1.161 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -1.415 C, which is 19.415 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.654 C, and the margin is measured, not chosen: 1.654 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -1.665 C, which is 19.665 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.606 C, and the margin is measured, not chosen: 1.606 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -1.665 C, which is 19.665 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.363 C, and the margin is measured, not chosen: 1.363 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -2.220 C, which is 20.220 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.059 C, and the margin is measured, not chosen: 1.059 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

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